Carbohydrates

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Carbohydrates

IIT-JEE Syllabus

Carbohydrates: Classification – mono, di and polysaccharides (Glucose, Sucrose and Starch only); hydrolysis of sucrose. Amino acids and Peptides: General structure and physical properties.

Objective

Subsequent to considering this unit, you will be ready to

- clarify the qualities of biomolecules like carbs, proteins and nucleic acids and chemicals;
- arrange starches, proteins, nucleic acids and nutrients on the premise of their constructions;
- clarify the contrast between DNA and RNA;
- depict the job of biomolecules in biosystem.

About Carbohydrate:

Old definition: $\rightarrow C_x(H_2O)_y$

Compounds follow $C_x(H_2O)_y$ general formula were generally considered as Carbohydrates

Limitations:

It was found by structural studies that carbohydrates did not contain intact water molecules. So these are not hydrates.

Example:

(i) $C_6H_{12}O_6 \rightarrow C_6(H_2O)_6$ ✓

(ii)  $\rightarrow C_2(H_2O)_2$ ×

(iii) $C_6H_{12}O_5 \rightarrow C_6(H_2O)_5H_2$ Carbohydrate ✓

(Rhamnose) Old definition not applicable

Definitions

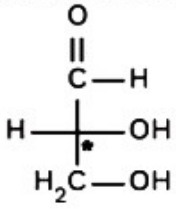
Initial studies based on molecular formulas suggested that carbon atoms are combined in some ways with water. Thus the name Carbohydrate was given.

Now carbohydrate includes:

- Polyhydroxy aldehydes
- Polyhydroxy ketones
- Compounds that are hydrolysed to them

Modern Definition:

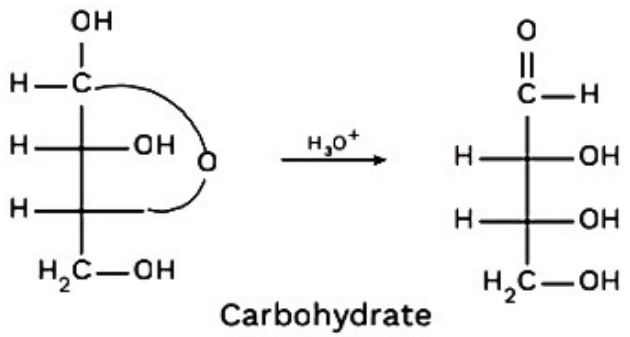
Polyhydroxy aldehyde or ketone having at least one chiral centre or which on hydrolysis gives polyhydroxy aldehyde or ketone are known as carbohydrates.



D – glyceraldehyde
(Smallest carbohydrate)



Hemiacetals or Hemiketals



Carbohydrate

Classification of Carbohydrate

Depending upon the number of simple carbohydrate molecules formed on hydrolysis, they are grouped into three classes:

(i) Monosaccharides:

They are simple carbohydrates.They can not be hydrolysed into simpler carbohydrates.

$$\underset{\text{glucose or fructose}}{C_6H_{12}O_6} + H_2O \xrightarrow{H^+} \text{No reaction}$$

(ii) Oligosaccharides:

The oligosaccharides (Oligo, Greek, few) are carbohydrates which yield a certain number (2-9) of monosaccharide molecules on hydrolysis.

(a) Disaccharides, : It gives 2 monosaccharide molecules on hydrolysis.

$$\underset{\text{sucrose}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{H^+} \underset{\text{glucose}}{C_6H_{12}O_6} + \underset{\text{fructose}}{C_6H_{12}O_6}$$

$$\underset{\text{Maltose}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{H^+} 2\,\underset{\text{glucose}}{C_6H_{12}O_6}$$

sucrose and maltose, equally of which have molecular formula, $C_{12}H_{22}O_{11}$

(b) Trisaccharides: It gives three monosaccharide molecules on hydrolysis

Points to remember

Formaldehyde (HCHO,CH_2O) and acetic acid [CH_3COOH, $C_2(H_2O)_2$] can be represented by the general formula, $C_x(H_2O)_y$ but they are not carbohydrates.

Note

- For hemiacetal formation: An aldehyde reacts with one molecule of an alcohol.
- For acetal formation: An aldehyde reacts with two molecules of an alcohol.
- Acetals are more stable than hemiacetals.

$$\underset{\text{raffinose}}{C_{18}H_{32}O_{16}} + 2H_2O \xrightarrow{H^+} \underset{\text{glucose}}{C_6H_{12}O_6} + \underset{\text{fructose}}{C_6H_{12}O_6} + \underset{\text{galactose}}{C_6H_{12}O_6}$$

(c) Tetra saccharides, which yield four nonosaccharide molecules on hydrolysis.

$$\underset{\text{Stachyose}}{C_{24}H_{32}O_{16}} \longrightarrow \underset{\text{Glucose}}{C_6H_{12}O_6} + \underset{\text{Fructose}}{C_6H_{12}O_6} + 2\underset{\text{Galactose}}{C_6H_{12}O_6}$$

(iii) Polysaccharides:

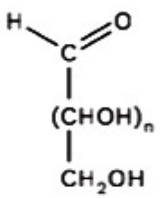
It gives more than 10 monosaccharide molecules on hydrolysis molecules on hydrolysis. Examples are cellulose and starch, both of which have molecular formula, $(C_6H_{10}O_5)_n$.

$$\underset{\text{starch}}{(C_6H_{10}O_5)_n} + nH_2O \xrightarrow{H^+} n\underset{\text{glucose}}{C_6H_{12}O_6}$$

Nomenclature :

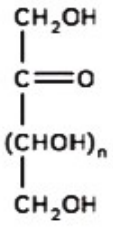
Number of C	Carbohydrates	Aldehyde	Ketone
3	Triose	Aldotriose	Ketotriose
4	Tetrose	Aldotetrose	Ketotetrose
5	Pentose	Aldopentose	Ketopentose
6	Hexose	Aldohexose	Ketohexose

The aldoses and ketoses may be characterized by the following general formulas.



Aldoses

(n = 1, 2, 3, 4, 5)

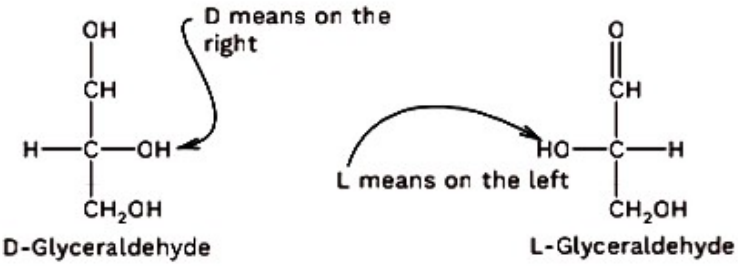


Ketoses

(n = 1, 2, 3, 4)

Glucose and fructose are particular examples of an aldose and a ketose.

D and L Terminology (Trioses)



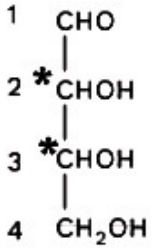
D-configuration: In Fischer projection, OH group is at right side on bottom asymmetric carbon.

L- configuration: In Fischer projection, OH group is at left side on bottom asymmetric carbon.

Naturally available sugars are mostly contain D-configuration.

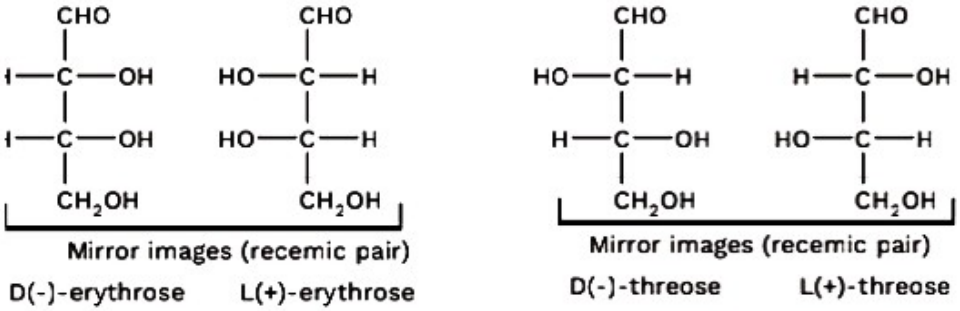
Aldotetroses

If we observe the general formula of an aldosterone, we see that they contain 2 asymmetric



an aldotetrose

This likely that 2^2 or 4 optical isomers are possible. They may be characterized as the following two pairs:



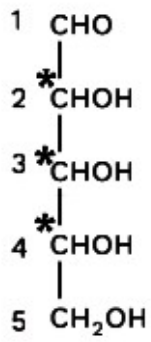
Note

Glyceraldehyde is the smallest aldose. It exists as a pair of enantiomers due to having an asymmetric carbon. The enantiomer with R-configuration was assigned as D-Glyceraldehyde. The mirror image of D- Glyceraldehyde is L-Glyceraldehyde.

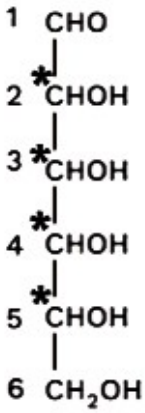
Points to remember

Just like R, S configuration, D and L represent the configuration of chiral centre. It has no relation with direction of rotation of plane polarised light.It is determined by the bottom asymmetric carbon. D and L are mirror images of each-other.

Aldopentoses and Aldohexoses



an aldopentose



an aldohexose

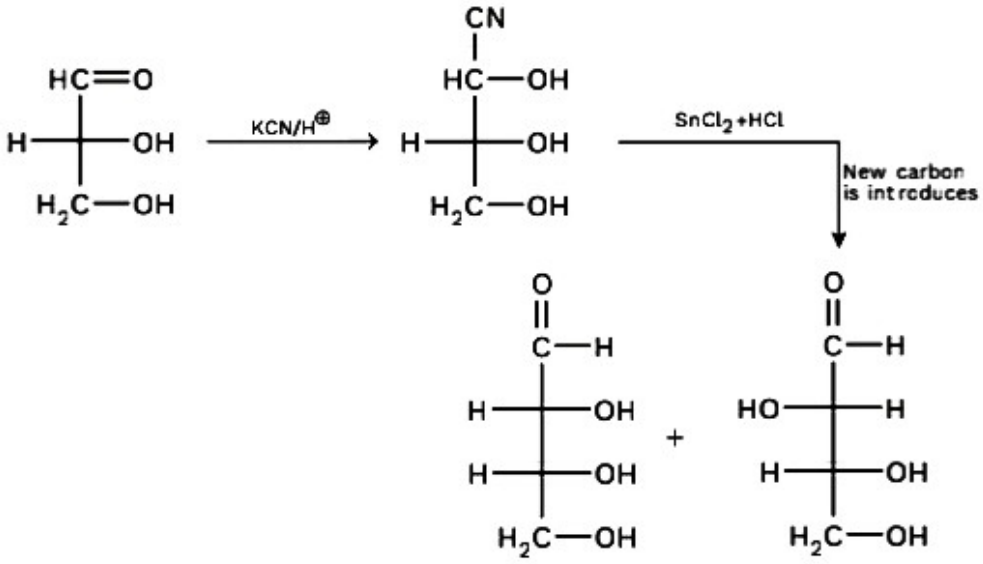
D-families Generation

Kiliani - Fischer Synthesis

Using this synthesis, the length of carbon chain of an aldose is increased by one carbon. It means we can synthesize pentose from tetrose and similarly hexose from pentose.

For example: When glyceraldehyde reacts with cyanide ion and HCl ,it forms a new chiral centre. It is due to the addition of cyanide ion to carbonyl carbon. Formation of this new asymmetric centre gives two cyanohydrin molecules . These two cyanohydrins are different only with respect to configuration at second carbon. The configurations at other asymmetric centres will remain same. These cyanohydrins can be converted to aldoses. This reaction forms a pair of C-2 epimers.

If we are taking D- glyceraldehyde as reactant, both the products will be D-sugars because the configuration at stereo centres remain same

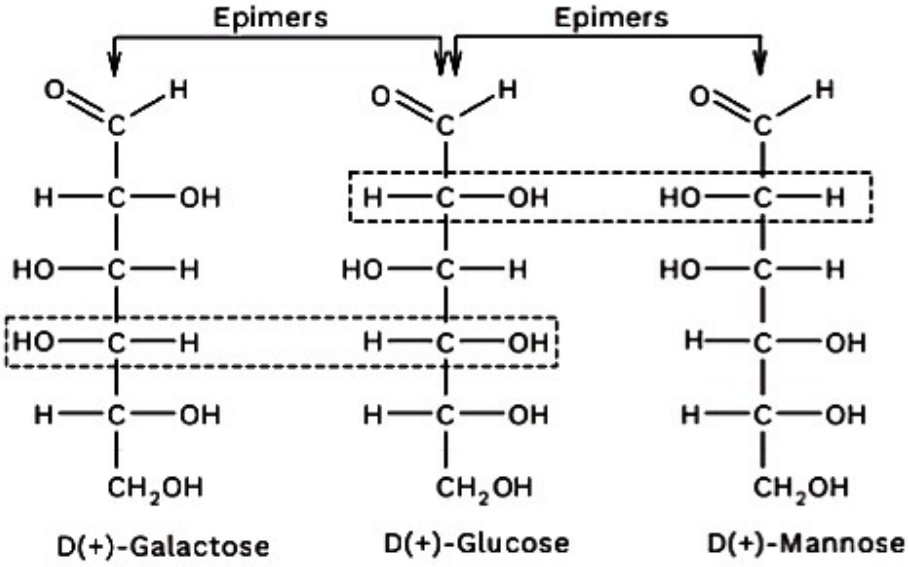


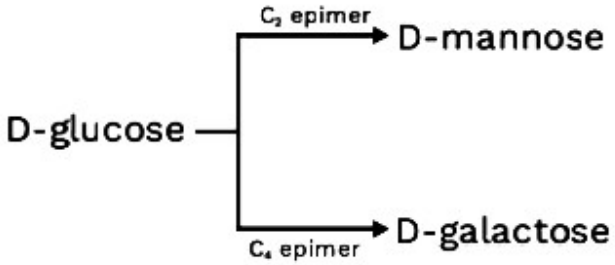
Epimers

These are diastereomers having different configuration at only one asymmetric carbon.

Example:

D(+)- glucose is epimeric with D(+) -mannose and D(+) -galactose as shown below:





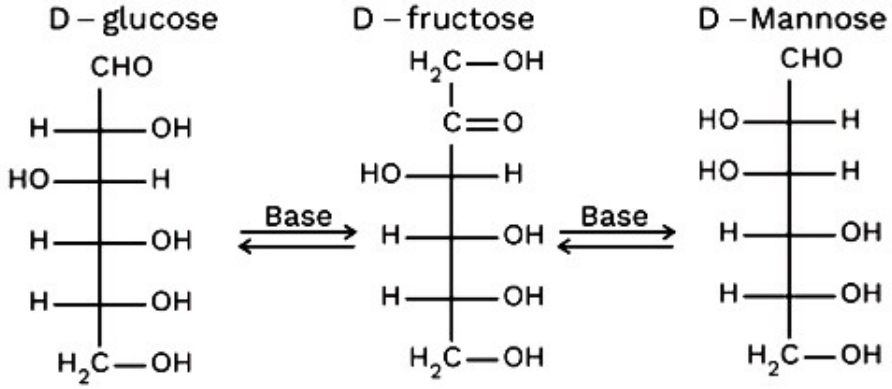
Points to remember

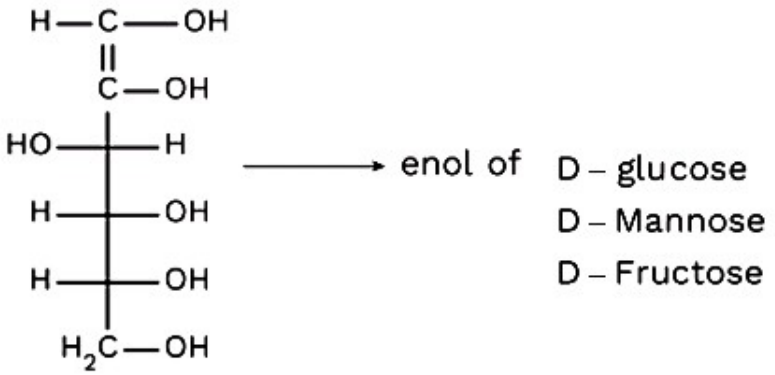
Epimers are diastereomers formed by changing configuration across 1 chiral centre only.

All epimers are diastereomers but reverse is not true.

Glyceraldehyde has no epimer D-erythro and D-threo are C_2 epimer

Lobry-De-Bruyn-Van-Ekenstein-Rearrangement





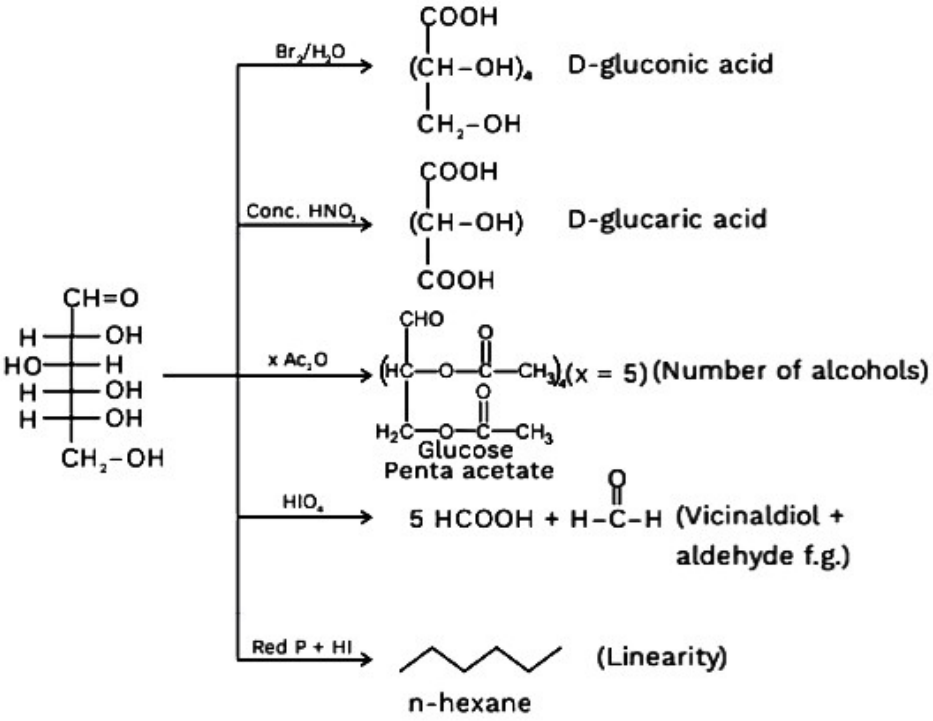
	D-glucose	D-mannose	D-fructos
Tollen's test	✓	✓	✓
Fehling	✓	✓	✓
Benedict	✓	✓	✓

Points to remember

Enol form of Glucose or Mannose will convert into Fructose in basic medium enolization process.

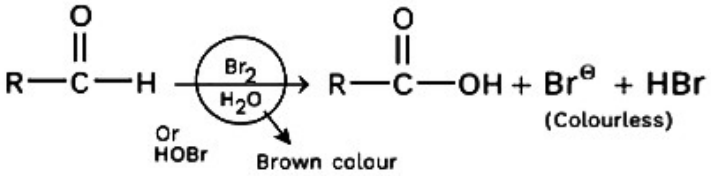
Fructose to Glucose or Mannose will convert during enol to keto formation.

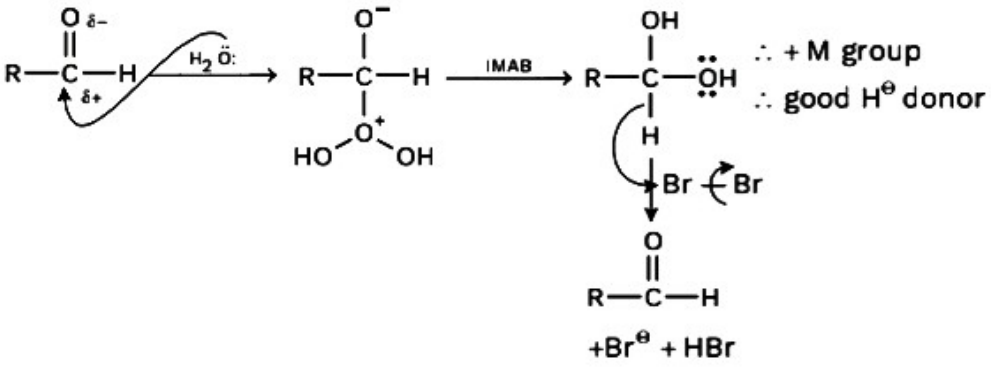
Reactions of Glucose



Reaction with Br_2/H_2O

It is a mild oxidising agent oxidises aldehyde only.



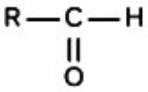


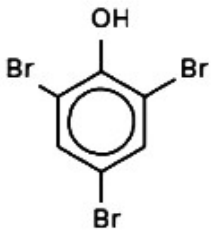
Practical Test	D-glucose	D-Frustose
Tollens test	✓	✓
Fehling test	✓	✓
Benedict test	✓	✓
Br_2/H_2O	✓	×

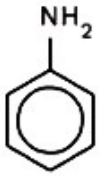
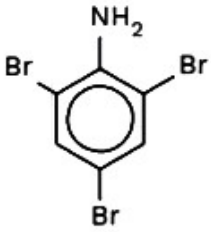
Bromine- Water Test(Br_2/H_2O)

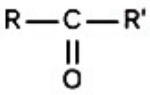
(i) C=C ✓

(ii) C ≡ C ✓

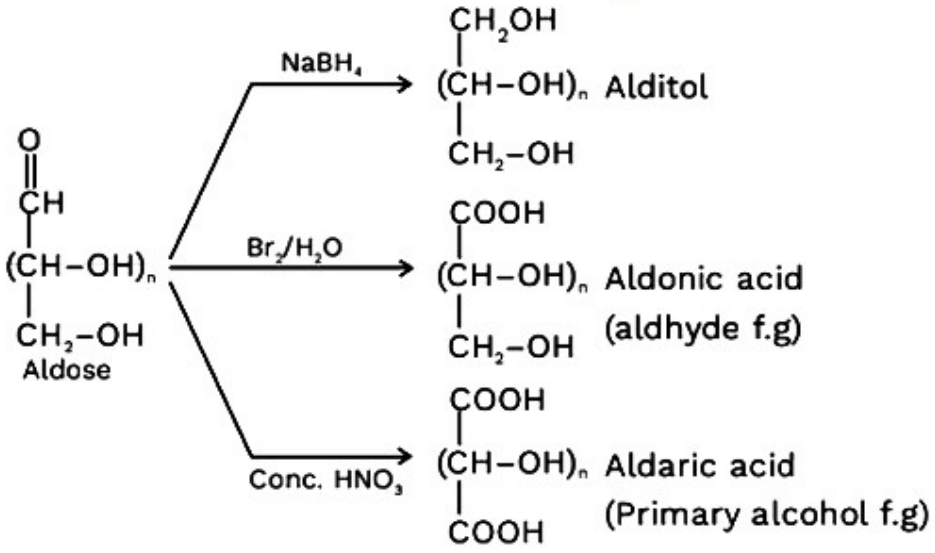
(iii)  ✓

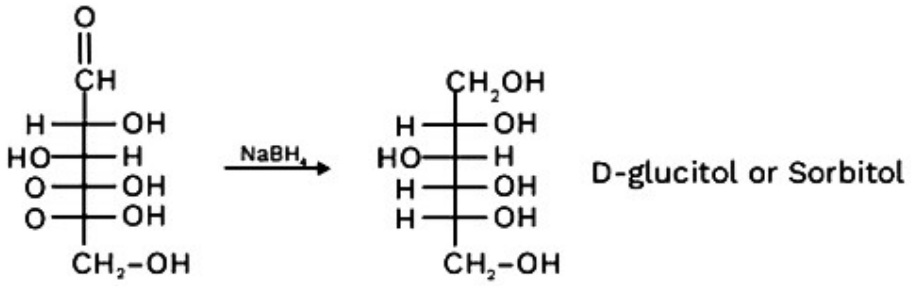
(iv)   White ppt ✓

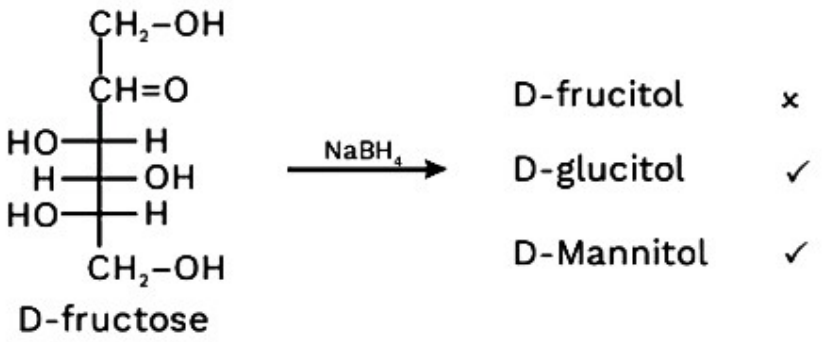
(v)   White ppt ✓

(vi)  ×

Oxidation and Reduction reaction of carbohydrates:







Due to equilibrium of fructose, mannose and glucose (C_2 epimer)

Reducing/Non-Reducing sugar

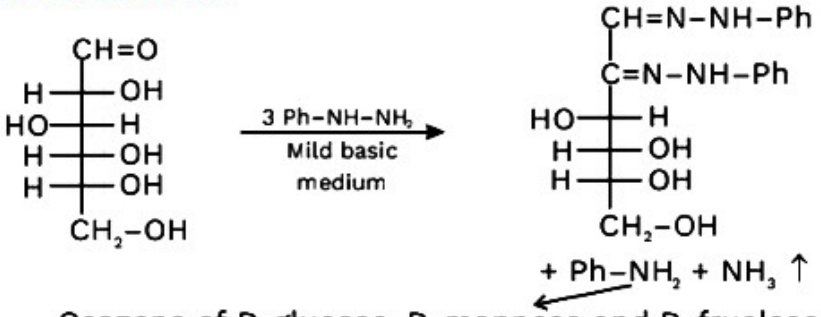
Reducing sugar are those carbohydrates which gives Tollen's, Fehling, Benedict test.

All monosaccharides are reducing sugar.

All those carbohydrates which do not give Tollen's, Fehling or benedict test are known non-reducing sugar.

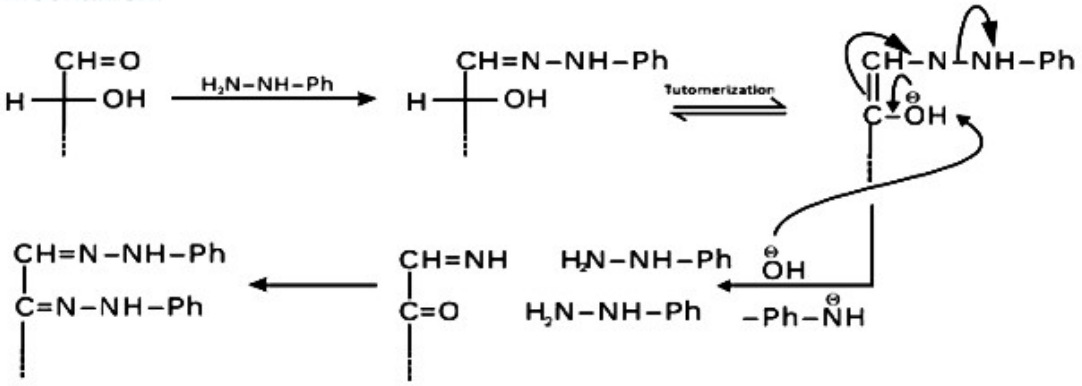
Example: Starch, cellulose, glycogen, sucrose, etc.

Osazone formation:



Osazone of D-glucose, D-mannose and D-fructose

Mechanism



Note:

It this reaction 3 moles of $Ph–NH–NH_2$ are consumed.

During this reaction C_2 loses its chirality.

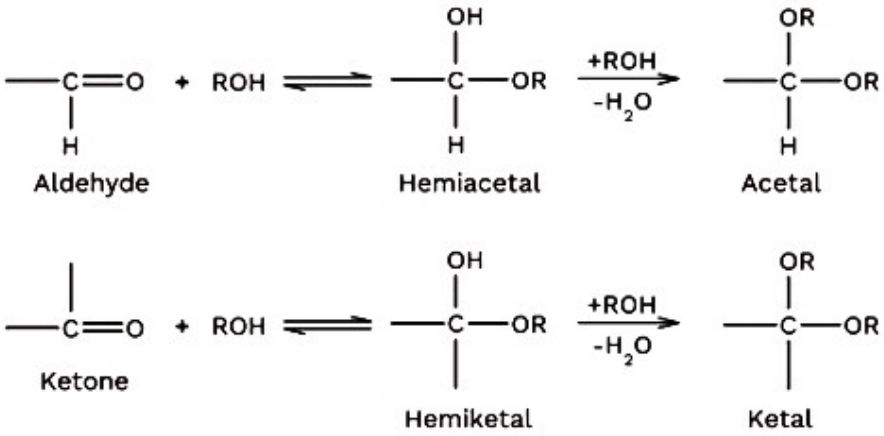
C_2 epimer will give same osazone.

D-mannose, D-fructose, D-glucose give same osozone formation due to ekenstein rearrangement.

Cyclic structure of monosaccharides

Cyclic hemiacetals

The reaction of aldose and ketose with alcohols can be summarised as,



Due to the presence of OH groups and carbonyl group , most of the carbohydrates are found as cyclic acetals or hemiacetals.

Since the OH group and carbonyl group are present in the same molecule, cyclic hemiacetal is formed.

If it results in five or six membered ring formation, it is considered as stable.

These are more stable than their open chain structure.

Hemiketal : It is cyclic hemiacetal of a ketone.

Pyranose: Six- membered cyclic hemiacetal

Furanose : Five-membered cyclic hemiacetal

Points to remember

Most important forms of a sugar:

(i) Open-chain structure

(ii) Five-membered oxygen heterocycle(Furanose)

(iii) Six-membered oxygen heterocycle(pyranose)



Glucose Exist in 3 different form:

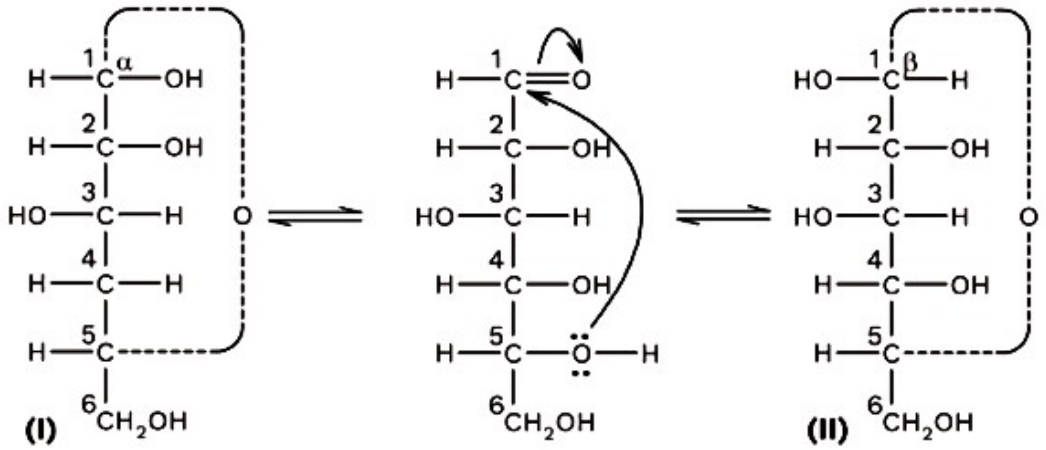
Following chemical property is explained by cyclic structure.

(i) Negative 2-4 DNP test

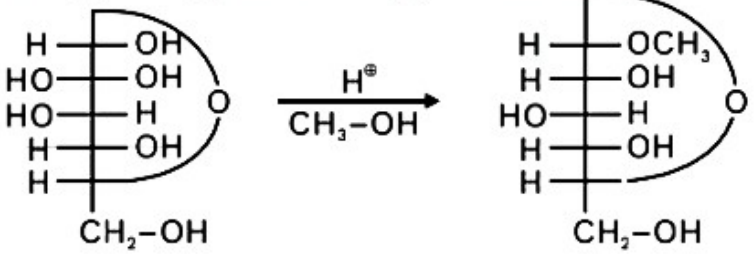
(ii) Negative Schiff's test

(iii) Negative Na_2S test

(iv) Cyclic structure of Glucose – Anomers

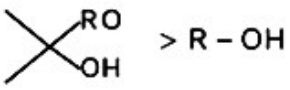


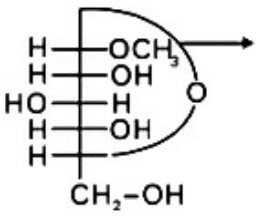
α–D(+)-Glucose or α–D(+)-Glucopyranose | D(+)-Glucose (Open chain form) | β–D(+)-Glucose or β–D(+)-Glucopyranose



It will not give tollen's test, not reducing sugar, mutarotation and ozone form.

	α-D-Glucopyranose	D-Glucose	β-D-glucose
Tollen's test	✓	✓	✓
Reducing sugar	✓	✓	✓
Mutarotation	✓	✓	✓
Osazone formation	✓	✓	✓

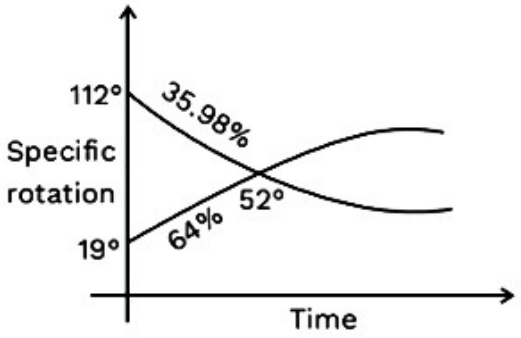


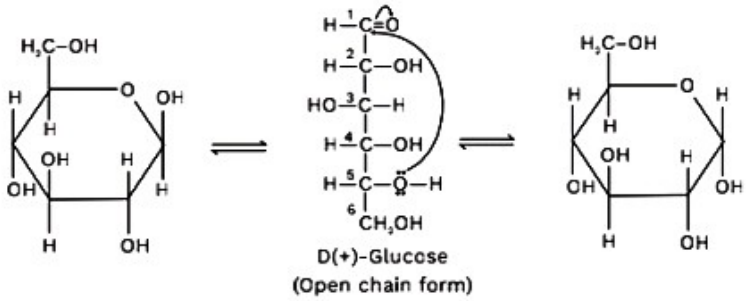


Glycoside (Acetal or ketal of carbohydrate)
* Mannoside (Acetal or ketal of carbohydrate)
* Fructoside (Acetal or ketal of carbohydrate)

Mutarotation:

The change in specific rotation of a n optically active compound in a solution w.r.t time to an equilibrium value is called mutarotaion (reactant should be in dil. base)





β-D-Glucopyranose	⟶ Anomers ⟷	α-D-Glucopyranose
64%	0.02%	35.98%
+18.9°		+112°
POS ×		POS ×
COS ✓		COS ×
Chiral ×		Chiral ✓
O.A. ✓		O.A. ✓

⇒ Condition for Mutarotation, Presence of hemiacetal group

Anomer:

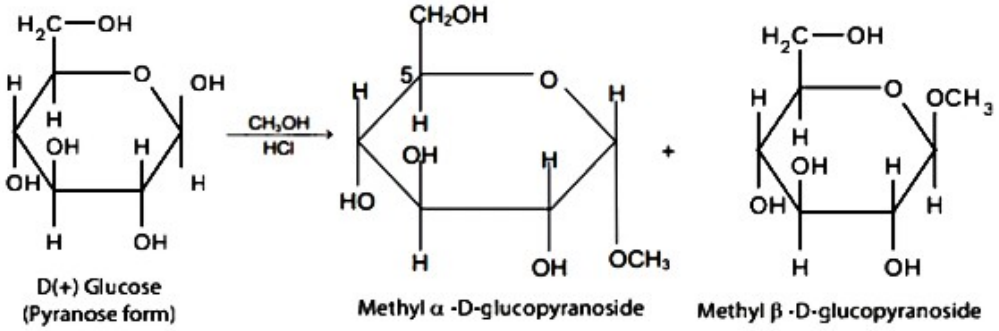
Cyclic epimers are known as anomers.

All anomers are epimers but vice – versa is not true.

In glucose, C_1 carbon is anomeric

In fructose, C_2 carbon is anomeric.

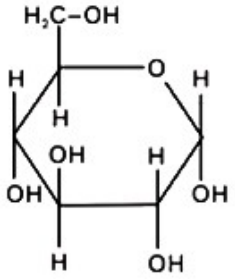
Glycoside formation: When a slight amount of gaseous HCl is passed into a solution of D (+) glucose in methanol , a reaction takes place that results in the creation of anomeric methyl acetals.



Carbohydrate acetals, normally are called glycosides and an acetal of glucose is called glucoside.

Haworth Projection:

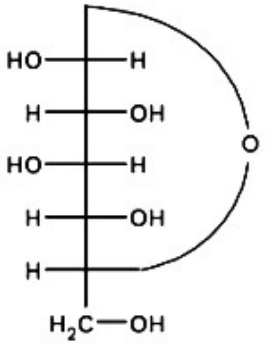
Given by Fischer.

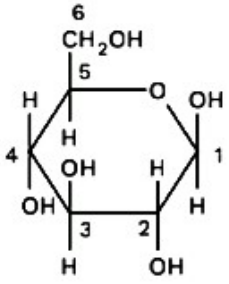


α-D-Glucopyranose

B	B	A	B
C_1	C_2	C_3	C_4

B = OH Down side . A= OH upside





β – D glucopyranose

β – D glucopyranose

A	B	A	B
C_1	C_2	C_3	C_4

B= OH Downside , A= OH upside

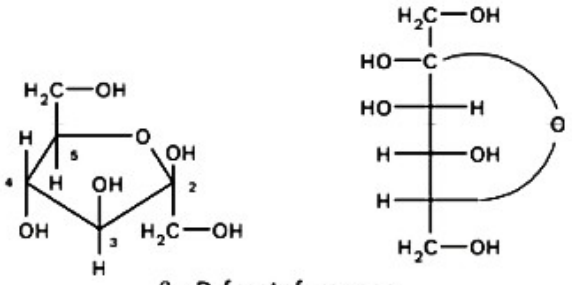
Rules for Haworth Projection:

Fischer to Haworth

- Draw hexagon with oxygen on top and right hand side.
- Number the carbon in clockwise direction.
- OH on right hand side below the plane.
- OH on left hand side above the plane.
- Sixth carbon is always above the plane.

Q1 **Draw the Haworth projection for Beta-D-fructofuranose**

Sol.



β - D fructofuranose

Disaccharides :

Carbohydrates which upon hydrolysis give 2 molecules of the same or different monosaccharides are called disaccharides. Their general formula is $C_{12}H_{22}O_{11}$.

The 3 most important disaccharides are maltose, sucrose and lactose.

Every one of these on hydrolysis with either an acid or an enzyme gives 2 molecules of the same or different monosaccharides as shown below:

$$\underset{\text{Sucrose}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow[\text{or invertase}]{H^+} \underset{\text{Glucose}}{C_6H_{12}O_6} + \underset{\text{Fructrose}}{C_6H_{12}O_6}$$

$$\underset{\text{Maltose}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow[\text{or Maltase}]{H^+} \underset{\text{Glucose}}{C_6H_{12}O_6} + \underset{\text{Glucose}}{C_6H_{12}O_6}$$

$$\underset{\text{Lactose}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow[\text{or Lactase}]{H^+} \underset{\text{Glucose}}{C_6H_{12}O_6} + \underset{\text{Galactose}}{C_6H_{12}O_6}$$

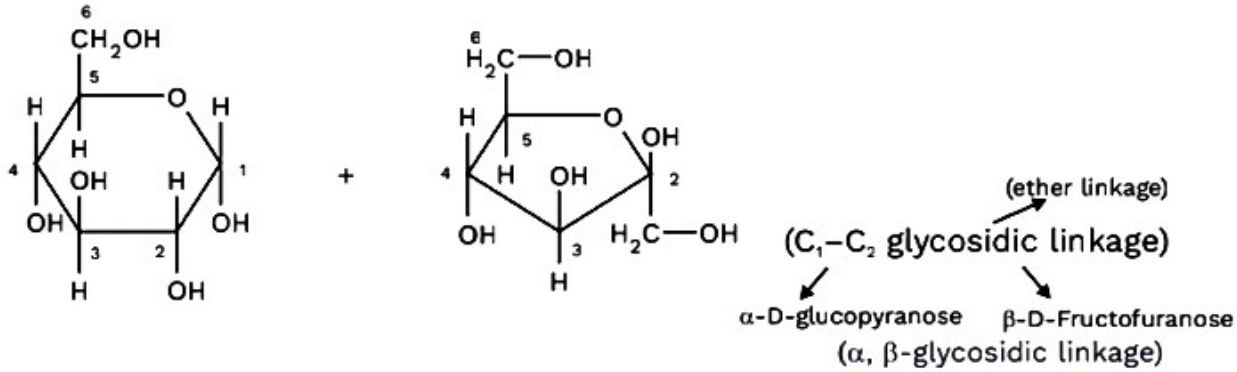
Points to remember

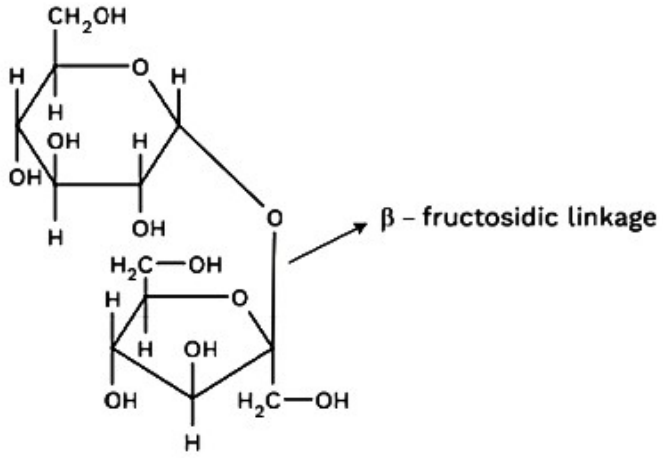
- Glycoside formation: The anomeric carbon reacts with the OH group
- Disaccharide: The glycoside product when hydroxy group is part of another sugar

Sucrose :

It is produced by condensation of one molecule of glucose and one molecule of fructose. Unlike lactose and maltose, it is non-reducing sugar since both glucose (C_1 - α) and fructose (C_2 - β) are attached to each other through their reducing centres. Its structure is shown below

1. Sucrose $\longrightarrow$ α-D-glucopyranose + β-D-fructofuranose
(C_1–C_2 linkage)





Hemiacetal	×	Because all are (+ve) due to presence of hemiacetal group
Tollen's test	×	
Reducing sugar	×	
Osozone	×	
Mutarotation	×	

Hydrolysis (Invert Sugar or Invertase):

Hydrolysis of sucrose with hot dilute acid yields D-fructose and D-glucose.

$$\underset{\text{Sucrose}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{H^+} \underset{\text{D-Glucose}}{C_6H_{12}O_6} + \underset{\text{D-Fructrose}}{C_6H_{12}O_6}$$

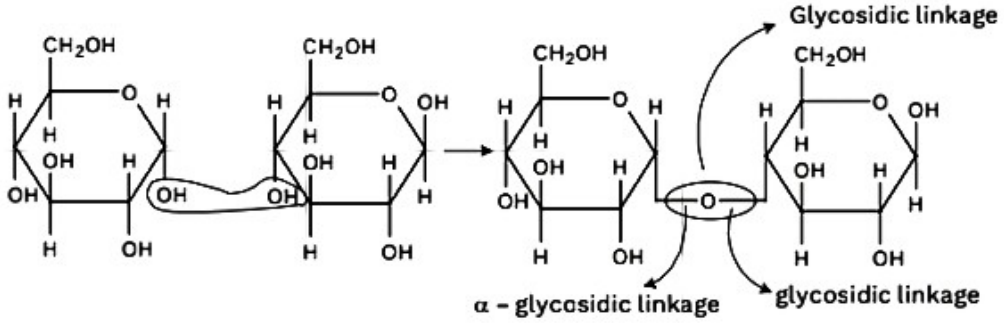
$[\alpha]_D = +66.5°$ $\qquad$ $[\alpha]_D = +53°\,[\alpha]_D = -92°$ (Invert Sugar)

$$[\alpha]_D = (+53°) - (-92°) = -39°\text{Suc}$$

dextrorotatory, its specific rotation being +66.5%, D glucose is also dextrorotatory, $[\alpha]_D = +53°$, but D-fructose has a large negative rotation, $[\alpha]_D = -92°$. Since D-fructose has a greater certain rotation than D glucose, the resulting mix is laevorotatory. For the reason that of this the hydrolysis of sucrose is called as the inversion of sucrose, and the equimolecular mixture of glucose and fructose is known is invert sugar or invertose

2. Maltose

α-D-glucopyranose + α-D-glucopyranose



Here no α-β linkage only α-glycosidic linkage because C_4 is not anomeric carbon.

Hemiacetal linkage	✓
Tollen's test	✓
Reducing Sugar	✓
Mutarotation	✓
Osazone formation	✓

Q2 **Why sucrose is inverted sugar?**

Sol. Sucrose is dextorotatory which on acidic hydrolysis becomes levorotatory.

Sucrose $\xrightarrow{\overset{\oplus}{O}H}$ no reaction

(+66.6°)

Sucrose $\xrightarrow{\overset{\oplus}{O}H}$ glucose + fructose

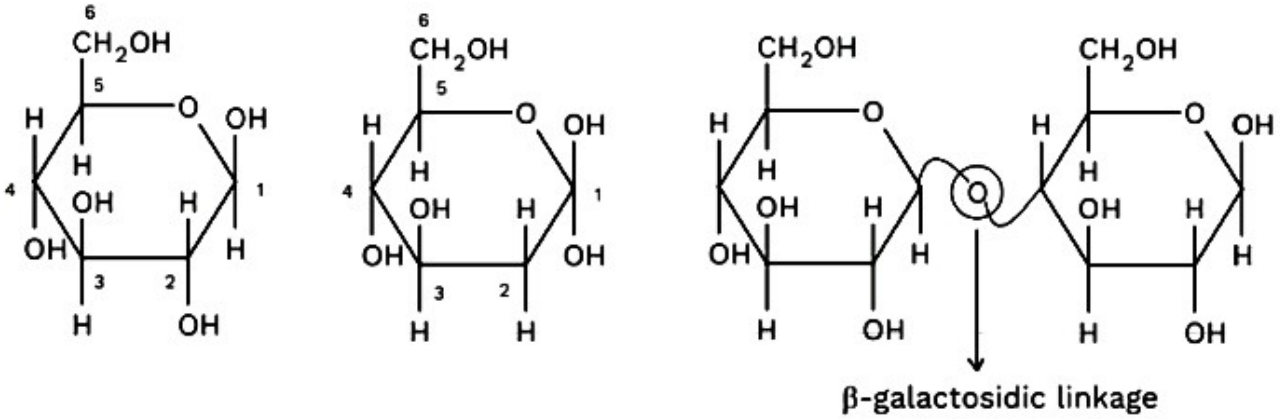
(+66.6°) (+52.4°) (−92.4°)

−40°

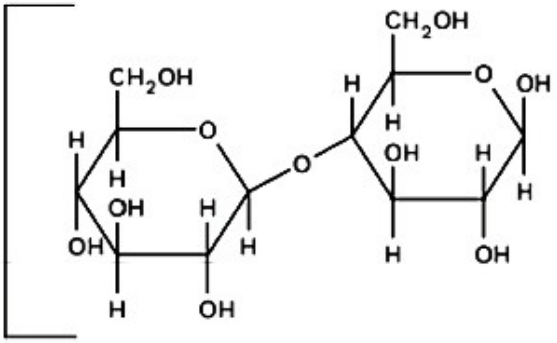
Lactose ⟶ galactose + glucose (C_1–C_4–glycosidic linkage)

β-D-galactopyranose β-D-glucopyranose

C_4 epimer



Hemiacetal	✓
Tollen's test	✓
Reducing sugar	✓
Osazone formation	✓
Mutarotation	✓



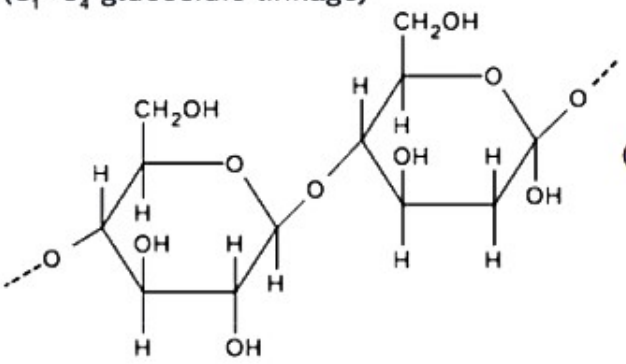
Sucrose is non-reducing, maltose and lactose are reducing sugar.

Polysaccharide

Polysaccharides are produced when a large number (hundreds to even thousands) of monosaccharide molecules join together with the elimination of water molecule. Thus, polysaccharides may be regarded as condensation polymers in which the monosaccharides are joined together by glycosidic linkages. Some important polysaccharides are:

1. Cellulose
2. Starch
3. Glycogen
4. Gums and
5. Pectins

1. Cellulose → Polymer of β-D-glucopyranose
(C_1–C_4 glucosidic linkage)



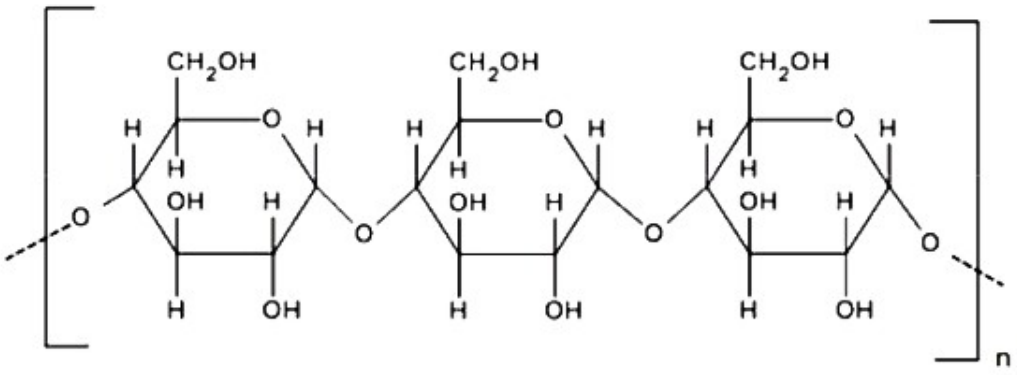
(Non–reducing sugar)

2. Starch →

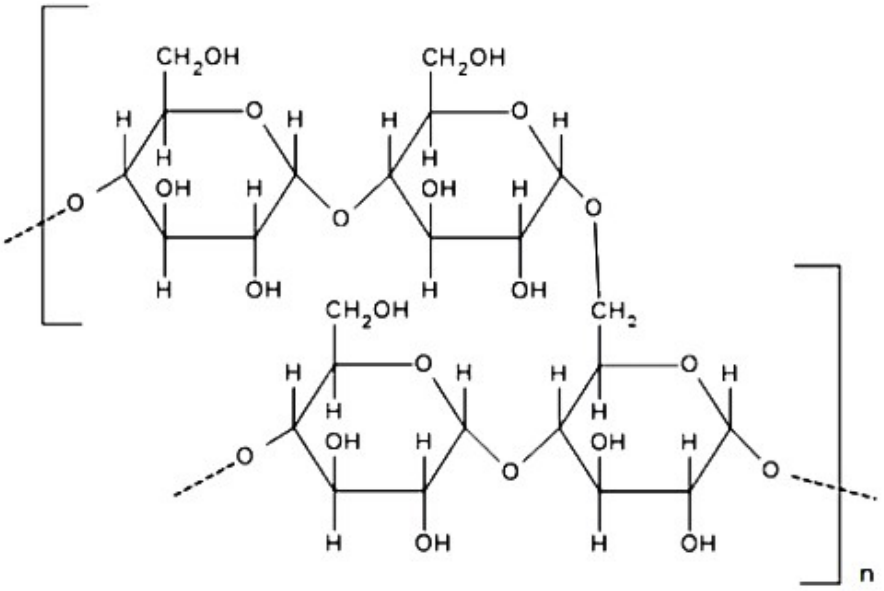
Polymer of α-D-glucopyrancal

Amylose	Amylopectin
* (15-20%)	* (80-85%)
* H_2O soluble	* H_2O insoluble
* C_1–C_4 glycotidic linkage	* C_1–C_4 and C_1-C_6 glucosidic linkage
* Linear structure	* Cross linked structure
* Non-reducing sugar	* Non-reducing sugar

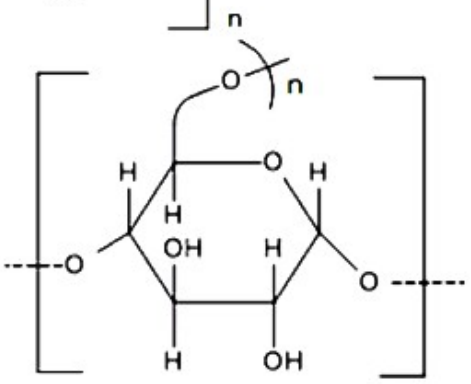
- **Amylose :**



- **Amylopectin :**



3. **Glycogen** → Polymer of α-D-glucopyranose (Animal storage) (C_1-C_4 and C_1-C_6) all
   All polysaccharides are non-reducing sugars because they do not contain hemiacetal.

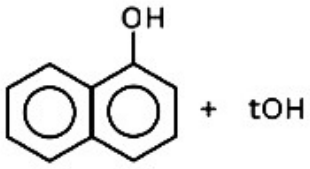


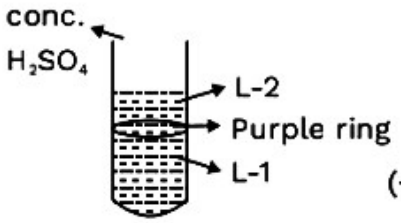
- **Test for Carbohydrates :** Glucose, Fructose, Sucrose etc.

  Molisch Test

  Molisch reagent :

  Alpha-napthol in ethanol solution





Carbohydrate + Molischreagent
(test solution)
Conc. H_2SO_4
(+ve test) Purple ring forms in between
L-1 and L-2 if carbohydrats